

Question 1 (20 marks)

- a) Find i) $\int \frac{x+4}{x^3+4x} dx$ 4
- ii) $\int \sec^2 x \sin x dx$ 3
- iii) $\int_0^{\sqrt{2}-1} \frac{dx}{\sqrt{3-x^2-2x}}$ 3
- b) Given that $I_n = \int_0^{\frac{\pi}{2}} \sin^n \theta d\theta$, show that $I_n = \frac{n-1}{n} I_{n-2}$.
Hence, evaluate $\int_0^{\frac{\pi}{2}} \sin^5 \theta d\theta$. 4
- c) Using the method of cylindrical shells, find the volume of the solid of revolution 3
formed when the region bounded by $y = 3x - x^2$ and $y = x$ is rotated
around the y-axis.
- d) The base of a certain solid is the circle $x^2 + y^2 = 9$. Cross-sections cut perpendicular 3
to the y-axis are squares with one side in the base. Find the volume of the solid.

Question 2 on page 2.

Question 2 (20 marks) Start this question on a new page.

- a) A particle is thrown vertically upward at a velocity of 60m/s. The retardation due to air resistance is proportional to the square of the velocity with the constant of proportionality being 0.01 (i.e. total resistance, $R = 0.01mv^2$).

Show that

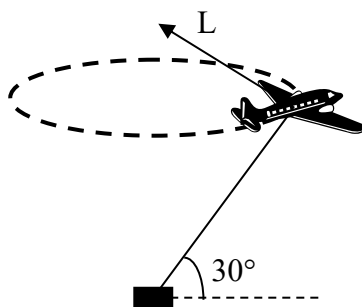
- i) the time taken for the particle to reach its maximum height is

given by $t = \frac{10}{\sqrt{g}} \tan^{-1}\left(\frac{6}{\sqrt{g}}\right)$ **3**

- ii) the maximum height reached is given by

$$h = 200 \ln\left(\frac{g + 36}{g}\right)$$
 3

- b) A model plane of mass 5kg attached to the end of an inelastic wire, of length 2m, whose other end is fixed, flies in a horizontal circle of elevation 30° . (Take $g = 9.8\text{ms}^{-2}$)



If the lift force, L , acts at right angles to the wire and L is twice the weight of the plane,

- i) Draw a diagram clearly showing the forces acting on the plane. **1**
- ii) Find the tension in the wire in Newtons, correct to 2 decimal places. **2**
- iii) Find the speed of the plane in m/s, correct to 1 decimal place. **2**
- c) A car of mass 1tonne travels in a level curve of radius 160m at a constant speed of 70km/h.
- i) Calculate the friction between the wheels and the road surface, correct to the nearest Newton. **2**
- ii) Assuming the road is banked at an angle of 10° , calculate the frictional force between the road surface and the wheels, correct to the nearest Newton (take $g = 9.8\text{ms}^{-2}$). **4**
- iii) If the road were banked at 10° , find the speed at which no frictional force is experienced, correct to the nearest m/s. **3**

END OF PAPER

EXT 2 ASSESS TERM 3 Solutions

QUESTION 1

a) i/ $\int \frac{x+4}{x^3+4x} dx = \int \frac{x+4}{x(x^2+4)} dx$

$$\frac{x+4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$x+4 = A(x^2+4) + x(Bx+C)$$

$$= Ax^2 + Bx^2 + Cx + 4A$$

$$\therefore A+B=0$$

$$C=1$$

$$4A=4$$

$$A=1 \therefore B=-1$$

$$\therefore \int \frac{x+4}{x^3+4x} dx = \int \frac{1}{x} - \frac{x-1}{x^2+4} dx$$

$$= \ln x - \frac{1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{1}{x^2+4} dx$$

$$= \ln x - \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

1- Partial fractions

1- split second numerator

1- correct ln primitive

1- correct $\tan^{-1}$ primitive

/4

ii/ I = $\int \sec^2 x \sin x dx$

$$\text{let } u = \sin x \quad v' = \sec^2 x$$

$$u' = \cos x \quad v = \tan x$$

$$\therefore I = \sin x \tan x - \int \cos x \tan x dx$$

$$= \sin x \tan x - \int \sin x dx$$

$$= \sin x \tan x + \cos x + C$$

or

$$= \int \sec x \tan x dx = \sec x + C$$

or

$$\int \sec^2 x \sin x dx = \int \frac{\sin x}{\cos^2 x} dx$$

$$= \frac{1}{\cos x} + C$$

$$\text{let } v = \cos x$$

$$v' = -\sin x$$

/3

1- separation into parts

1- uses correct formula

1- correct answer

or
1- rearrange
1- balance
1- correct answer

iii/ $\int_0^{\sqrt{2}-1} \frac{dx}{\sqrt{3-x^2-2x}} = \int_0^{\sqrt{2}-1} \frac{dx}{\sqrt{4-(x+1)^2}}$

$$= \left[\sin^{-1} \left(\frac{x+1}{2} \right) \right]_0^{\sqrt{2}-1}$$

$$= \left(\sin^{-1} \frac{\sqrt{2}}{2} - \sin^{-1} \frac{1}{2} \right)$$

$$= \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$$

$$= \frac{\pi}{12}$$

/3

1- complete square in denominator

1- correct $\sin^{-1}$ primitive

1- correct evaluation

$$b) I_n = \int_0^{\pi/2} \sin x \sin^{n-1} x dx$$

$$\text{Let } u = \sin^{n-1} x \quad v' = \sin x$$

$$u' = (n-1) \sin^{n-2} x \cos x \quad v = -\cos x$$

$$I_n = \left[-\cos x \sin^{n-1} x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^2 x \sin^{n-2} x dx$$

$$= 0 - (-1) \sin^{n-1} x \Big|_0^{\pi/2} + (n-1) \int_0^{\pi/2} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$= (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

Hence,

$$\begin{aligned} \int_0^{\pi/2} x dx &= I_5 \\ &= \frac{4}{5} I_3 \\ &= \frac{4}{5} \cdot \frac{2}{3} I_1 \\ &= \frac{8}{15} \int_0^{\pi/2} \sin x dx \\ &= \frac{8}{15} [-\cos x]_0^{\pi/2} \\ &= \frac{8}{15} \end{aligned}$$

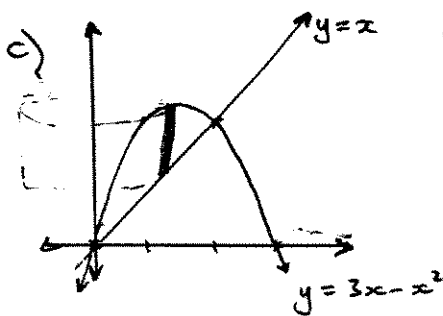
1- Integ by parts

1- simplification to given form

1- uses reduction formula

1- correct answer

/4



Intersection

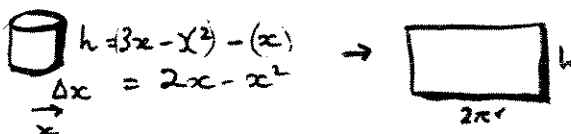
$$x = 3x - x^2$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$\therefore x=0, x=2$$

One Cylinder



$$\Delta V = 2\pi x (2x - x^2) \Delta x$$

$$\therefore V = 2\pi \int_0^2 (2x^2 - x^3) dx$$

$$= \pi \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2$$

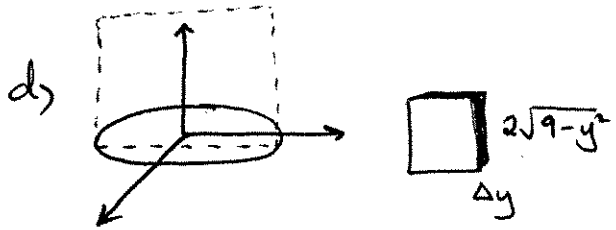
$$= \frac{8\pi}{3} u^3$$

1- finds intersection

1- ΔV

1- correct volume

/3



$$\Delta v = (2\sqrt{9-y^2})^2 \Delta y$$

$$v = 4 \int_{-3}^3 9 - y^2 dy$$

$$= 8 \left[9y - \frac{y^3}{3} \right]_0^3$$

$$= \underline{144 u^3}$$

1/3

1- side length

1- Δv w.r.t. y .

1- correct volume.

$$(2) (a) \ddot{x} = \frac{-v^2 - 100g}{100}$$

① a correct expression for $\ddot{x}$

$$\frac{dt}{dv} = \frac{-100}{v^2 + 100g}$$

$$\int_0^T dt = \int_{60}^0 \frac{-100 dv}{v^2 + 100g}$$

$$T = 100 \int_0^{60} \frac{dv}{v^2 + 100g}$$

① Time = a correct definite integral

$$= \frac{100}{10\sqrt{g}} \left[\tan^{-1} \frac{v}{10\sqrt{g}} \right]_0^{60}$$

① Correct integration leading to the answer

$$= \frac{10}{\sqrt{g}} \tan^{-1} \frac{6}{\sqrt{g}}$$

Note: If done as an indefinite integral
2nd mark is for evaluating the constant.

$$(b) v \frac{dv}{dx} = \frac{-v^2 - 100g}{100}$$

①

$$\frac{dx}{dv} = \frac{-100v}{v^2 + 100g}$$

$$\int_0^H dx = \int_{60}^0 \frac{-100v}{v^2 + 100g} dv$$

①

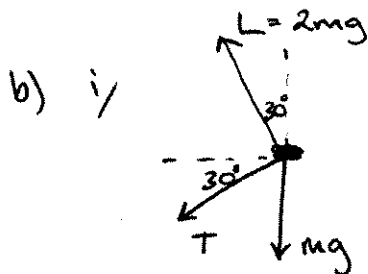
$$H = 50 \int_0^{60} \frac{2v}{v^2 + 100g} dv$$

$$= 50 \left[\log(v^2 + 100g) \right]_0^{60}$$

①

$$= 50 \log \left(\frac{36 + g}{g} \right)$$

no penalty.



1 - all forces labelled.

/1

ii/ Vertically: $L \cos 30^\circ - mg - T \sin 30^\circ = 0$

$$2mg \cdot \frac{\sqrt{3}}{2} - mg = \frac{1}{2} T$$

$$T = 2mg (\sqrt{3} - 1)$$

$$= 2 \times 5 \times 9.8 (\sqrt{3} - 1)$$

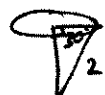
$$= \underline{71.74 \text{ N}}$$

/2

1 - vertical components

1 - answer

iii/ Horizontally: $\frac{mv^2}{r} = L \sin 30^\circ + T \cos 30^\circ$



$$r = 2 \cos 30^\circ$$

$$= \sqrt{3}$$

$$v^2 = \frac{r}{m} (2mg \cdot \frac{1}{2} + T \cdot \frac{\sqrt{3}}{2})$$

$$v^2 = 38.496$$

$$v = \underline{6.2 \text{ m/s}}$$

/2

1 - horizontal components

1 - answer

c) i/ $m = 1000$

$$v = \frac{70}{3.6} \text{ m/s}$$

$$r = 160 \text{ m}$$

$$F = \frac{mv^2}{r}$$

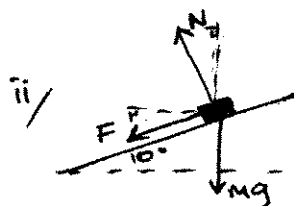
$$= \frac{1000 \times \left(\frac{70}{3.6}\right)^2}{160}$$

$$= \underline{2363 \text{ N}}$$

/2

1 - formula

1 - answer



Horizontally: $\frac{mv^2}{r} = F \cos 10^\circ + N \sin 10^\circ$ ①

Vertically: $N \cos 10^\circ - F \sin 10^\circ - mg = 0$ ②

$$\therefore N = \frac{F \sin 10^\circ + mg}{\cos 10^\circ}$$

Into ① $\frac{mv^2}{r} = F \cos 10^\circ + \tan 10^\circ (F \sin 10^\circ + mg)$

$$F (\cos 10^\circ + \tan 10^\circ \sin 10^\circ) = \frac{mv^2}{r} - mg \tan 10^\circ$$

$$F = \frac{\frac{mv^2}{r} - mg \tan 10^\circ}{\cos 10^\circ + \tan 10^\circ \sin 10^\circ}$$

$$= \underline{625 \text{ N}}$$

/4

1 - horiz.
1 - components of force

1 - vert. components

1 - correct rearrangement

1 - answer

iii/ $F=0$

Vertically: $N \cos 10^\circ - mg = 0$

$$N = \frac{mg}{\cos 10^\circ}$$

Horizontally: $\frac{mv^2}{r} = N \sin 10^\circ$

$$\therefore \frac{mv^2}{r} = mg \tan 10^\circ$$

$$\therefore v^2 = rg \tan 10^\circ$$

$$v = \underline{32 \text{ m/s}}$$

1- state $F=0$

1- resolution of components

1- answer.

QUESTION 2

a) $\uparrow$ $\downarrow$ $\downarrow$ $\downarrow$
 mg R

$$F = -mg - R$$

$$ma = -mg - 0.01mv^2$$

$$a = -g - 0.01v^2$$

i) $a(v) \rightarrow v(t)$ (Max ht when $v=0$)

$$a = \frac{dv}{dt}$$

$$\frac{dv}{dt} = -g - 0.01v^2$$

$$\frac{dt}{dv} = \frac{-1}{g + 0.01v^2}$$

$$t = - \int_{60}^0 \frac{1}{g + 0.01v^2} dv$$

$$= \left[\frac{1}{0.01} \cdot \frac{1}{\sqrt{g \cdot 0.01}} \tan^{-1} \left(\frac{v}{\sqrt{g \cdot 0.01}} \right) \right]_0^{60}$$

$$= \frac{\sqrt{0.01}}{\sqrt{g}} \cdot \frac{1}{0.01} \tan^{-1} \left(\frac{60 \times 0.1}{\sqrt{g}} \right)$$

$$= \frac{10}{\sqrt{g}} \tan^{-1} \left(\frac{6}{\sqrt{g}} \right)$$

/3

1- use $a = \frac{dv}{dt}$

1- correct integral

1- rearrange

ii) $a(v) \rightarrow v(x)$

$$a = v \frac{dv}{dx}$$

$$v \frac{dv}{dx} = -g - 0.01v^2$$

$$\frac{dv}{dx} = \frac{-g - 0.01v^2}{v}$$

$$\frac{dx}{dv} = \frac{-v}{g + 0.01v^2}$$

$$x = -200 \int_{60}^0 \frac{0.02v}{g + 0.01v^2} dv$$

$$= 200 [\ln(g + 0.01v^2)]_0^{60}$$

$$= 200 \ln \left(\frac{g + 36}{g} \right)$$

/3

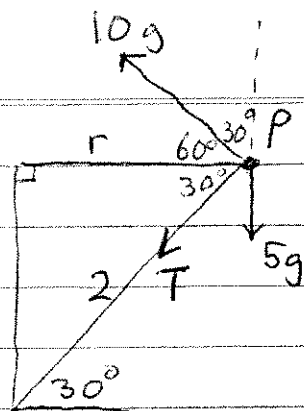
1- use $a = v \frac{dv}{dx}$

1- correct integral

1- rearrange

(b) (i)

$$r = 2 \cos 30^\circ \\ = \sqrt{3}$$



① forces

(ii) Vertical forces at P

$$10g \cos 30^\circ = T \sin 30^\circ + 5g$$

① 3 vertical forces

$$T = 10g(\sqrt{3} - 1) \\ \approx 71.74 \text{ N}$$

①

(iii) Horizontal forces at P

$$10g \sin 30^\circ + T \cos 30^\circ = \frac{5v^2}{r}$$

① ≤ 2 forces = $\frac{mv^2}{r}$

$$5g + \frac{T\sqrt{3}}{2} = \frac{5v^2}{\sqrt{3}}$$

$$v^2 = \frac{\sqrt{3}}{5} \left(5g + \frac{T\sqrt{3}}{2} \right)$$

$$v \approx 6.2 \text{ m s}^{-1}$$

①

$$(c) m = 1000 \text{ kg}$$

$$r = 160 \text{ m}$$

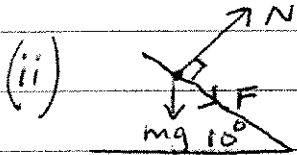
$$v = \frac{70 \times 1000}{60 \times 60}$$

$$= \frac{175}{9} \text{ ms}^{-1}$$

$$(i) F = \frac{mv^2}{r} \quad (1)$$

$$= \frac{1000 \times \left(\frac{175}{9}\right)^2}{160}$$

$$\approx 2363 \text{ N} \quad (1)$$



Horizontal

$$N \sin 10^\circ + F \cos 10^\circ = \frac{mv^2}{r} \quad (1)$$

Vertical

$$N \cos 10^\circ - F \sin 10^\circ = mg \quad (2)$$

$$(1) \times \cos 10^\circ - 2 \times \sin 10^\circ \quad (1)$$

$$F(\cos^2 10^\circ + \sin^2 10^\circ) = \frac{mv^2}{r} \cos 10^\circ - mg \sin 10^\circ$$

$$F \approx 625 \text{ N} \quad (1)$$

$$(iii) F = 0$$

$$\therefore N \sin 10^\circ = \frac{mv^2}{r} \quad (3)$$

$$N \cos 10^\circ = mg \quad (4)$$

$$(3) \div (4) \quad \tan 10^\circ = \frac{v^2}{gr} \quad (1)$$

$$v^2 = gr \tan 10^\circ, v \approx 17 \text{ ms}^{-1} \quad (1)$$